

PROGRAM NO:01

```
#include <iostream>

using namespace std;

int calculate2Low(int arr[], int size)
{
    int lowest = arr[0];
    for (int i = 1; i < size; i++)
    {
        if (arr[i] < lowest)
        {
            lowest = arr[i];
        }
    }

    int secondLowest = 1000000;
    for (int i = 0; i < size; i++)
    {
        if (arr[i] > lowest && arr[i] < secondLowest)
        {
            secondLowest = arr[i];
        }
    }
}
```

```
}
```

```
    return secondLowest;
```

```
}
```

```
int calculate2High(int arr[], int size)
```

```
{
```

```
    int highest = arr[0];
```

```
    for (int i = 1; i < size; i++)
```

```
    {
```

```
        if (arr[i] > highest)
```

```
        {
```

```
            highest = arr[i];
```

```
        }
```

```
    }
```

```
    int secondHighest = -1000000;
```

```
    for (int i = 0; i < size; i++)
```

```
    {
```

```
        if (arr[i] < highest && arr[i] > secondHighest)
```

```
        {
```

```
            secondHighest = arr[i];
```

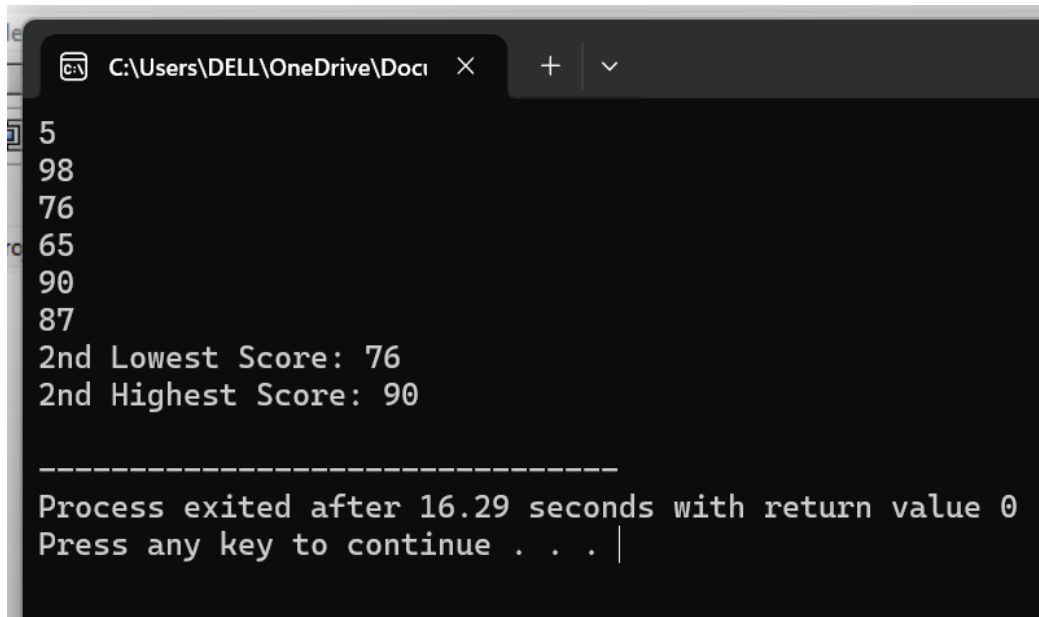
```
    }  
}
```

```
    return secondHighest;  
}
```

```
int main()  
{  
    int N;  
  
    cin >> N;  
  
    int scores[N];  
  
    for (int i = 0; i < N; i++)  
    {  
        cin >> scores[i];  
    }  
  
    cout << "2nd Lowest Score: " << calculate2Low(scores,  
N) << endl;
```

```
        cout << "2nd Highest Score: " << calculate2High(scores,  
N) << endl;
```

```
    return 0;  
}
```



```
5  
98  
76  
65  
90  
87  
2nd Lowest Score: 76  
2nd Highest Score: 90  
  
-----  
Process exited after 16.29 seconds with return value 0  
Press any key to continue . . . |
```

PROGRAM NO:02

```
#include <iostream>
```

```
#include <iomanip>
```

```
using namespace std;
```

```
double calculateSum(int arr[], int size)
```

```
{  
    int sum = 0;  
  
    for (int i = 0; i < size; i++)  
    {  
        sum += arr[i];  
    }  
  
    return sum;  
}
```

```
int main()  
{  
    int N;  
    cin >> N;  
  
    int scores[N];  
  
    for (int i = 0; i < N; i++)  
    {  
        cin >> scores[i];  
    }  
}
```

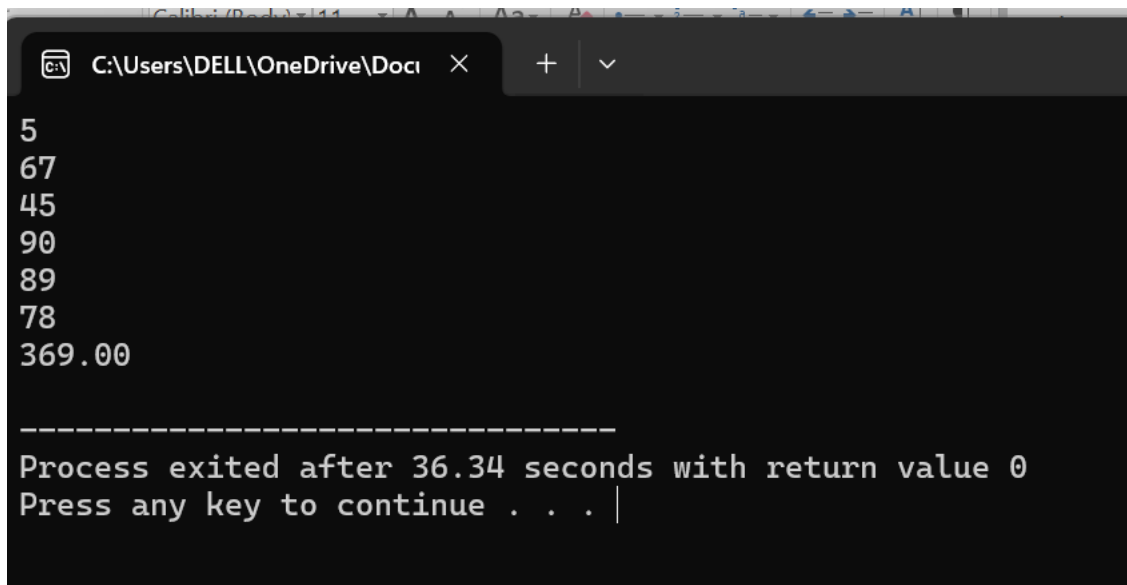
```
}
```

```
double total = calculateSum(scores, N);
```

```
cout << fixed << setprecision(2) << total << endl;
```

```
return 0;
```

```
}
```



```
C:\Users\DELL\OneDrive\Docu  X + v
5
67
45
90
89
78
369.00

-----
Process exited after 36.34 seconds with return value 0
Press any key to continue . . . |
```

PROGRAM NO:03

```
#include <iostream>
```

```
#include <iomanip>
```

```
using namespace std;
```

```
double celsiusToFahrenheit(double celsius)
```

```
{  
    return (celsius * 9.0 / 5.0) + 32;  
}
```

```
int main()
```

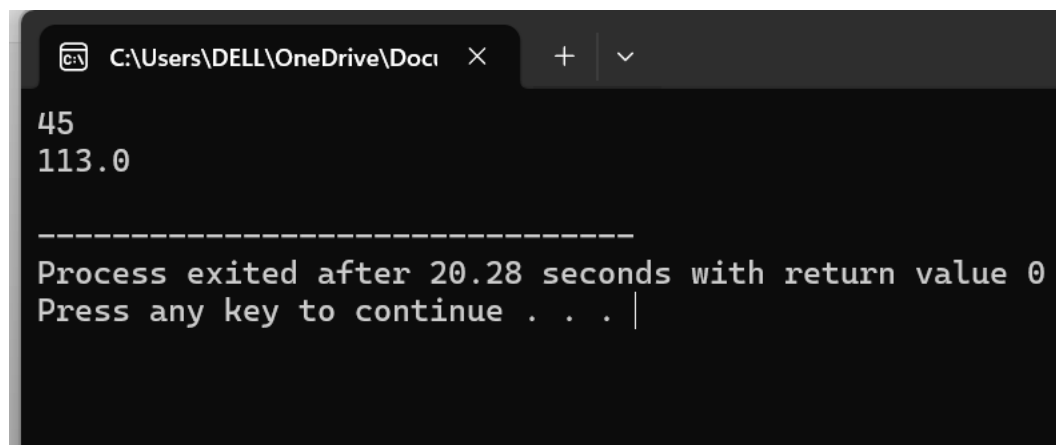
```
{  
    double celsius;
```

```
    cin >> celsius;
```

```
    double fahrenheit = celsiusToFahrenheit(celsius);
```

```
    cout << fixed << setprecision(1) << fahrenheit << endl;
```

```
    return 0;}
```



```
C:\Users\DELL\OneDrive\Docu  X  +  v  
45  
113.0  
-----  
Process exited after 20.28 seconds with return value 0  
Press any key to continue . . . |
```

PROGRAM NO:04

```
#include <iostream>
```

```
using namespace std;
```

```
bool isLeapYear(int year) {
```

```
    if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0))  
{
```

```
        return true;
```

```
    }
```

```
    return false;
```

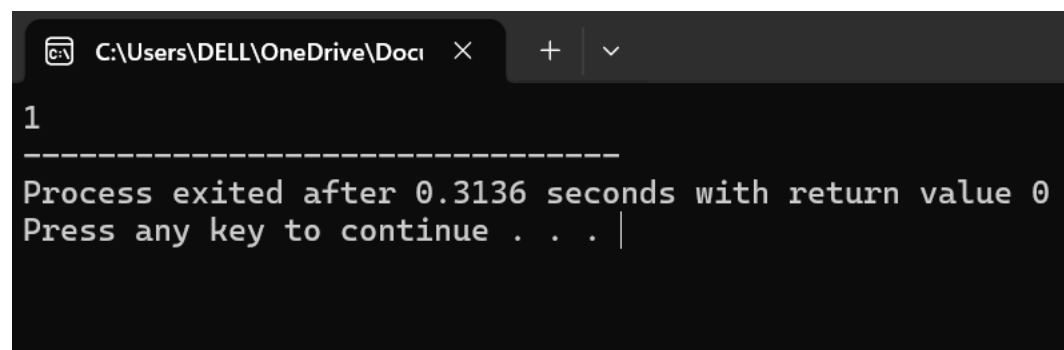
```
}
```

```
int main() {
```

```
    cout << isLeapYear(2024);
```

```
    return 0;
```

```
}
```

A screenshot of a Windows command prompt window. The title bar shows the file path 'C:\Users\DELL\OneDrive\Docu' and a close button. The terminal content shows the number '1' on the first line, followed by a dashed line separator. Below the separator, it displays the output of the program: 'Process exited after 0.3136 seconds with return value 0' and 'Press any key to continue . . . |' with a cursor at the end.

```
1  
-----  
Process exited after 0.3136 seconds with return value 0  
Press any key to continue . . . |
```


PROGRAM NO:05

```
#include <iostream>
```

```
using namespace std;
```

```
int countVowels(char word[])
```

```
{
```

```
    int count = 0;
```

```
    for (int i = 0; word[i] != '\0'; i++)
```

```
    {
```

```
        if (word[i] == 'a' || word[i] == 'e' || word[i] == 'i' ||
```

```
            word[i] == 'o' || word[i] == 'u' ||
```

```
            word[i] == 'A' || word[i] == 'E' || word[i] == 'I' ||
```

```
            word[i] == 'O' || word[i] == 'U')
```

```
        {
```

```
            count++;
```

```
        }
```

```
    }
```

```
    return count;
```

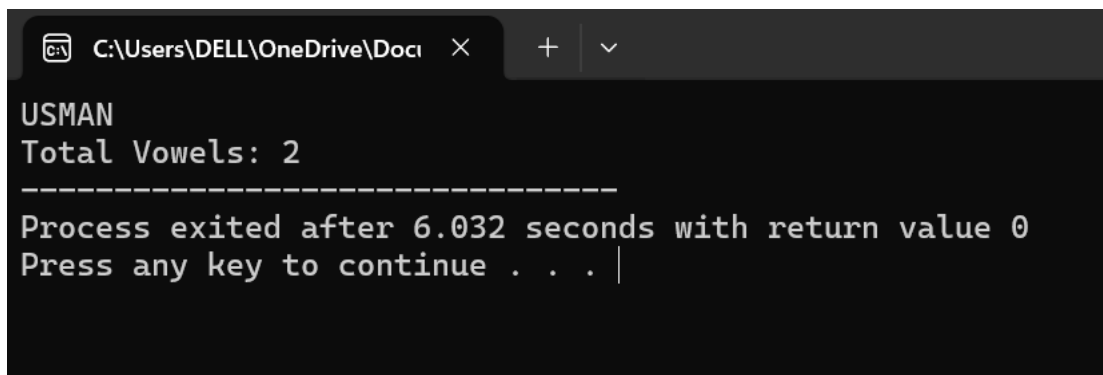
```
}
```

```
int main()
{
    char name[100];

    cin >> name;

    cout << "Total Vowels: " << countVowels(name);

    return 0;
}
```



```
C:\Users\DELL\OneDrive\Docu  ×  +  ∨
USMAN
Total Vowels: 2
-----
Process exited after 6.032 seconds with return value 0
Press any key to continue . . . |
```

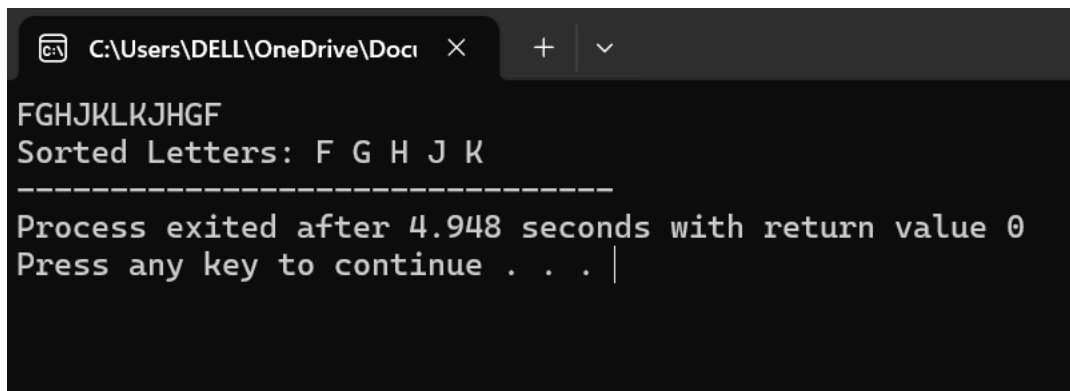
PROGRAM NO:06

```
#include <iostream>
using namespace std;
```

```
int main()
{
    char arr[5];
    char temp;

    for (int i = 0; i < 5; i++)
    {
        cin >> arr[i];
    }
    for (int i = 0; i < 4; i++)
    {
        for (int j = 0; j < 4 - i; j++)
        {
            if (arr[j] > arr[j + 1])
            {
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
}
```

```
    cout << "Sorted Letters: ";  
    for (int i = 0; i < 5; i++)  
    {  
        cout << arr[i] << " ";  
    }  
  
    return 0;  
}
```



The screenshot shows a Windows command prompt window with a dark background. The title bar at the top indicates the file path 'C:\Users\DELL\OneDrive\Docu' and has standard window controls. The command prompt displays the following text: 'FGHJKLKJHGF' on the first line, 'Sorted Letters: F G H J K' on the second line, a dashed line '-----' on the third line, 'Process exited after 4.948 seconds with return value 0' on the fourth line, and 'Press any key to continue . . .' on the fifth line, followed by a cursor '|'.

```
FGHJKLKJHGF  
Sorted Letters: F G H J K  
-----  
Process exited after 4.948 seconds with return value 0  
Press any key to continue . . . |
```